

UPDATE BRIEF II FOR AN ORDER ON THE LEECH LATTICE FROM A $\mathbb{Z}_3$ -SYMMETRIC TRIPLE-OCTONION PRODUCT HISTORICAL ACCURACY OF THE KIRMSE–COXETER CORRECTION

JENS KÖPLINGER

ABSTRACT. The manuscript version of 29 April 2026 (v3) is frozen while under review. This is the *second*, separate update brief for that frozen version; it is not to be merged with the first (`update_brief_2026-04-29_v3.tex`, which covers a footnote correction and an appendix simplification). The present brief concerns one topic only: how the manuscript describes the Kirmse–Coxeter correction of the integral octonions (the “Kirmse twist”) — in the Introduction, in the two relevant Remarks of Sections 2 and 3, and in the Section 6 historical paragraph. **Its contents are provisional.** They are to be finalised only after the primary sources are obtained and the history is re-derived from scratch. Detailed reasoning and verbatim source quotations are in `paper/reviews/2026-05-17_claude_opus_4_7_kirmse_history_note.md`.

1. PLAN

The work proceeds in two steps, in order:

- (1) **Verify the history from primary sources.** Obtain the original papers (Section 2), read them, and derive the historical account from scratch — confirming what is correct and identifying any errors, in the manuscript and in the project’s own reference files. No secondary source (Petersson, Conway–Smith, Wikipedia, Baez) is to be taken on trust; they serve only as cross-checks.
- (2) **Elevate the wording.** Once the history is settled, rewrite the description of the Kirmse twist. The current wording — “swapping two basis indices in the Fano-plane labelling”, “there are seven of them” — is imprecise and is to be replaced by an accurate, structurally explicit account.

Everything in Sections 2–3 below is *pending* step (1). Nothing here is final.

2. PRIMARY SOURCES TO OBTAIN

The mathematical facts in this area (E_8 is a maximal order; there are seven; the twist yields a closed order) are to be verified *computationally* in-project, not cited. Only the *historical* record requires the original documents:

- **Kirmse (1924)** — *required*. J. Kirmse, “Über die Darstellbarkeit natürlicher ganzer Zahlen als Summen von acht Quadraten . . .,” Ber. Verh. Sächs. Akad. Wiss. Leipzig, Math.-Phys. Kl. **76** (1924), 63–82. German. Settles whether Kirmse proposed one lattice or seven.
- **Coxeter (1946)** — *required*. H.S.M. Coxeter, “Integral Cayley numbers,” Duke Math. J. **13** (1946), 561–578. Settles the precise form of the correction and how Bruck is credited. Paywalled online; needs a library.
- **Dickson (1923)** — *recommended; already free online* (Numdam). L.E. Dickson, “A new simple theory of hypercomplex integers,” J. Math. Pures Appl. (9) **2** (1923), 281–326.

- **Mahler (1942)** — *optional, not load-bearing*. K. Mahler, “On ideals in the Cayley–Dickson algebra,” Proc. Roy. Irish Acad. Sect. A **48** (1942), 123–133.

There is no separate Bruck publication; his correction is recorded within Coxeter (1946).

3. ITEMS PROVISIONALLY IDENTIFIED

The following are the candidate changes, **each pending verification** in step (1). They are recorded here so the digest has a starting point.

- **Credit Bruck.** The secondary record (Petersson 2018) attributes the actual correction to R. H. Bruck, with Coxeter publishing it. The manuscript credits only Coxeter. To be confirmed against Coxeter (1946) and then corrected in the prose.
- **The correcting operation.** The secondary record describes it as interchanging the identity element with one imaginary unit (seven choices $\rightarrow$ seven maximal orders), *not* as “swapping two basis indices in the Fano-plane labelling.” The accurate description — and its precise relation to the manuscript’s transposition σ of two *imaginary* units — is to be established from Coxeter (1946).
- **Re-examine the Remark contrasting σ with a Fano-line permutation.** It currently asserts that Coxeter’s correction is “not induced by a linear map on $\mathbb{R}^8$.” If the correction is a coordinate interchange, that assertion is wrong and the intended contrast must be restated. To be settled from Coxeter (1946).
- **Did Kirmse propose seven orders?** The secondary sources disagree. This is the central open historical question and is to be decided from Kirmse (1924) itself, cross-checked with Coxeter (1946).
- **Dickson’s priority.** Dickson (1923) reportedly gave a correct integral-octonion construction more than twenty years before Coxeter. Whether, and how, to record this (and to cite Dickson, and optionally Mahler) is to be decided after reading Dickson (1923).
- **Project reference files.** `background_octonions.txt` and `terminology.md` are to be checked for the same issues, including a discrepancy in the name of Kirmse’s journal.

NOTE ON THE COMPANION DOCUMENT

`companion.tex` carries the same imprecise wording. It is a frozen training document and is not edited here, but the same correction should be applied if the companion is ever revised.